Chapter 1

(Not-So-)Quick Users Guide

The purpose of this document is to provide a "quick" introduction to the fundamental objects in `oomph-lib`. Assuming that you

- have a basic understanding of the finite element method

and

- are reasonably familiar with object-oriented programming in C++,

the Quick Guide should enable you to solve basic PDE problems using `oomph-lib`. You should also consult the extensive `list of example codes` distributed with the library.

1.1 How to solve simple PDEs with oomph-lib

To solve a given PDE with `oomph-lib`, its discretisation needs to be specified in a `Problem` object. This usually involves the specification of the `Mesh` and the element types, followed by the application of the boundary conditions. If these steps are performed in the `Problem` constructor, the driver code itself can be as simple as:

```
int main()
 {
  //Build the problem
  DemoPoissonProblem problem;

  //Solve the problem, using Newton's method
  problem.newton_solve();

 }
```

The amount of work required to represent a given PDE as a `Problem` object is problem-dependent:

- If `oomph-lib's` existing `Mesh` objects and element types are sufficient to discretise the problem, the generation of a fully-functional `Problem` object is straightforward. We shall discuss this case in the section How to build a Problem.
- If `oomph-lib` provides the required element type but none of the available meshes are suitable, you will have to write your own `Mesh` object. We shall discuss this in the section How to build a Mesh. It is also possible to create `oomph-lib` meshes based on the output from third-party (commercial) mesh generators. This is illustrated in `another example.`

- Finally, the problem might involve a PDE for which `oomph-lib` does not provide any suitable elements. In this case you will need to write your own element. When implementing elements in `oomph-lib` the element formulation is generally sub-divided into (at least) two levels. Geometric elements implement the element geometry (e.g. 2D quad elements, 3D brick elements, etc); these geometric elements act as base classes for specific `FiniteElements` that implement the discretisation of particular PDEs. We discuss the implementation of the two types of elements in two separate sections:
  - In the section How to implement a new system of equations as a specific FiniteElement we describe how to discretise a new system of equations, using existing geometric elements.
  - In the section How to build a new geometric element we discuss how to formulate new geometric elements.

In all cases, we start with a brief description of the minimum functionality that must be implemented for the various objects. The general principles are then illustrated by considering the discretisation of the model problem

One-dimensional model Poisson problem

Solve

$$\frac{\mathrm{d}^2 u}{\mathrm{d}x^2} = \pm 30 \sin(\sqrt{30}x), \qquad (1)$$

in the one-dimensional domain $x \in [0, 1]$, subject to

$$u(0) = 0 \quad \text{and} \quad u(1) = \mp 1, \qquad (2)$$

by isoparametric two-node elements. (Please consult the `Introduction` for a review of the relevant finite element theory.) The source code for this example (`one_d_poisson_generic_only.cc`) is self-contained and uses only objects from the "generic" part of the `oomph-lib` library. [`oomph-lib` does, of course, provide 1D meshes and Poisson elements but we deliberately ignore these because we wish to illustrate how to build such objects from scratch. The alternative example code `one_d_poisson.cc`, discussed in `another example` illustrates the implementation of the same problem using existing library objects.]

The main purpose of this document is to provide a *"quick"* introduction to `oomph-lib`'s fundamental objects, so the example classes presented below are fully-functional but very basic. We provide exercises at the end of each section to allow the reader to explore various straightforward improvements, most of which are implemented in the corresponding objects in the library. Finally, the section Further comments provides a discussion of some general design principles that should be respected when creating new classes for use with the library.

1.2 How to build a Problem

All specific `Problems` should be implemented as objects that inherit from the generic `Problem` class. These specific `Problem` objects will vary considerably depending on the exact details of the problem being solved.
Only a few member functions *must* be implemented for each specific `Problem` class:

- The problem constructor
  `SomeSpecificProblem(...)`
- and the member functions
  `Problem::actions_before_newton_solve()`
  `Problem::actions_after_newton_solve()`

1.2.1 The constructor: SomeSpecificProblem(...)

The Problem constructor `SomeSpecificProblem(...)` usually contains the following steps:

- Construction of the timestepper (if required), e.g.
```
Problem::add_time_stepper_pt(new SomeTimeStepper());
```

- Construction of the `Mesh` (which may involve passing a particular element type as a template parameter), e.g.
```
Problem::mesh_pt() = new SomeMesh<SomeElement>(...);
```

- Specification of the boundary conditions. By default, all nodal values are assumed to be free and their values are initialised to zero. Therefore, we only need to "pin" the values that are prescribed by boundary conditions and assign any nonzero boundary values. For instance, the following instruction "pins" the (single) nodal value at node "0":
```
Problem::mesh_pt()->node_pt(0)->pin(0);
```
A non-zero boundary value can be set by
```
Problem::mesh_pt()->node_pt(0)->set_value(0,1.0);
```
This statement sets the first (= zero-th in C++'s zero-based indexing) nodal value to 1.0.

- Completion of the build of the elements: Elements are typically constructed during the mesh generation process. To allow a generic implementation of this process, mesh constructors use an argument-free constructor to create the elements. Thus, element constructors must not have any arguments. (See the section How to build a completely new Mesh , below, for an illustration of this process.) If an element requires any global parameters (such as pointers to physical parameters, function pointers to source functions, etc.), these parameters should be passed to the element via suitable access functions, *after* the element has been created by the mesh constructor.

- Assignment of the global and local equation numbers
```
Problem::assign_eqn_numbers();
```

1.2.2 The function Problem::actions_before_newton_solve()

`oomph-lib` treats all problems as non-linear problems and employs Newton's method to solve the system of nonlinear algebraic equations that results from its spatial (and, in time-dependent problems, temporal) discretisation. Within this framework, linear problems are special cases for which Newton's method converges in one step.
The (pure virtual) member function `Problem::actions_before_newton_solve()` should contain everything that *must* be done before a nonlinear solve to complete the specification of the problem. For example, the function may contain a call to update the boundary conditions.
```
void actions_before_newton_solve() {update_boundary_conditions();}
```
If desired, finer granularity may be obtained by overloading the empty virtual functions `Problem::actions_before_newton_convergence_check()`, `Problem::actions_before_newton_step()` and/or `Problem::actions_before_implicit_timestep()`, which are executed before each calculation of the maximum value of the residuals, each Newton step or each timestep, respectively. We refer to a separate document for a more detailed discussion of the various "action functions" executed by `oomph-lib's` Newton solver.

1.2.3 The function Problem::actions_after_newton_solve()

The (pure virtual) member function `Problem::actions_after_newton_solve()` should contain everything that is to take place after each nonlinear solve. For example, the function might contain commands to update the values of any "dependent" variables, or post-processing commands.
```
void actions_after_newton_solve()
 {
  //Update value of the slave variable in namespace GlobalVariables
  GlobalVariables::slave = node_pt(0)->value(0) + node_pt(0)->value(1);

  //Call the output function
  ofstream output_file("result.dat");
  output(output_file);
 }
```
Again, the finer-grained member functions `Problem::actions_after_newton_step()` and `Problem::actions_after_implicit_timestep()` are provided.

1.2.4 An example of a "complete" specific Problem

Here is the full specification for the `DemoPoissonProblem` class (taken from `one_d_poisson_generic_only.cc`) which implements the discretisation of the 1D Poisson problem described above, using a `Mesh` of type `OneDimMesh`, with elements of type `TwoNodePoissonElements`.

```
class DemoPoissonProblem : public Problem
 {
  public:

  /// Problem constructor: Pass the sign of the source function (default
  /// is +1)
  DemoPoissonProblem(const int& sign=1) : Sign(sign)
   {
    //Create a OneDimMesh Mesh object and set it to be the problem's mesh.
    //The element type, TwoNodePoissonElement, is passed  as a template
    //parameter to the mesh. The argument to the constructor indicates
    //the number of elements in the mesh.
    Problem::mesh_pt() = new OneDimMesh<TwoNodePoissonElement>(10);

    //Pin the unknowns at the ends of the 1D domain:
    //The 1D mesh has 2 boundaries, each of which contains a single node;
    //the nodes on the boundary are available from Mesh::boundary_node_pt(...)

    //Pin the single nodal value at the single node on mesh
    //boundary 0 (= the left domain boundary at x=0)
    mesh_pt()->boundary_node_pt(0,0)->pin(0);

    //Pin the single nodal value at the single node on mesh
    //boundary 1 (= the right domain boundary at x=1)
    mesh_pt()->boundary_node_pt(1,0)->pin(0);

    // All values are initialised to zero. This is consistent with the
    // boundary condition at x=0 and no further action is required
    // at that node.

    // Apply the boundary condition at x=1: u(x=1)=-/+1
    mesh_pt()->boundary_node_pt(1,0)->set_value(0,-double(Sign));


    // Finish problem setup: Set the sign for the source function
    // in all elements

    //Find number of elements in mesh
    unsigned n_element = mesh_pt()->nelement();

    // Loop over the elements
    for(unsigned i=0;i<n_element;i++)
     {
      // The sign() member function is defined in the
      // TwoNodePoissonElement class not the base GeneralisedElement class.
      // In order to use it, we must
      // upcast from GeneralisedElement to the specific element type,
      // which is achieved by a C++ dynamic_cast.
      TwoNodePoissonElement *specific_element_pt
       = dynamic_cast<TwoNodePoissonElement*>(mesh_pt()->element_pt(i));

      // Set the sign of the source function
      specific_element_pt->sign() = Sign;
     }

    //Assign the global and local equations numbers for the problem
    cout « "Number of equations is " « assign_eqn_numbers() « std::endl;
   }


  /// Check that everything has been set up properly
  void actions_before_newton_solve()
   {
    if (0==self_test())
     {
      cout « "Problem has been set up correctly and can be solved."
           « std::endl;
     }
    else
     {
      throw
       OomphLibError("Trouble! Check error messages and fix the problems.\n",
                     "DemoPoissonProblem::actions_before_newton_solve()",
                     OOMPH_EXCEPTION_LOCATION);
     }
   }


  /// Print out the result after the solve
  void actions_after_newton_solve()
   {
```

```
    ofstream file("result.dat");
    mesh_pt()->output(file);
   }

 private:

  /// The sign of the source function
  int Sign;

}; //End of problem definition
```

1.2.5 Exercises:

- Compile and run the example code one_d_poisson_generic_only.cc and compare the numerical results against the (fish-shaped) exact solution

$$u_{fish}(x) = \pm \left[\left(\sin(\sqrt{30}) - 1 \right) x - \sin(\sqrt{30}x) \right].$$

  (The sign of the source function is an optional argument to the problem constructor. If no argument is specified, the + sign is used.) Modify the `actions_after_newton_solve()` function so that it writes the exact solution into another file, "exact_solution.dat", say.

- Vary the number of elements and observe how the numerical results converge to the exact solution.
- Adjust the boundary conditions:
  - Change the boundary condition at the left end of the domain to $u(0) = 1$.
  - What happens when you don't apply a boundary condition at $x = 1$? (Hint: What are the natural boundary conditions for the Poisson equation?)
  - What happens when no boundary conditions are applied?
- Suppress the assignment of the equation numbers in the problem constructor and observe what happens. Check the error messages produced by `Problem::self_test()`.

1.3 How to build a Mesh

`oomph-lib` provides a large number of fully-functional `Mesh` objects. Many of these meshes can easily be adapted to discretise other domains, provided the domain has the same topology as the original `Mesh`. We shall illustrate this in section How to adapt an existing Mesh to a different domain shape below.

1.3.1 How to build a completely new Mesh

If the domain is too different from any of the existing meshes, a new, customised `Mesh` object, `SpecificMesh`, say, must be created. The specific mesh should be created as an object that inherits from the generic `Mesh` class. To maximise the potential for code-reuse, we recommended making the element type a template parameter of the `SpecificMesh` class.

```
template<class ELEMENT>
class SpecificMesh : public Mesh
```

This allows the `SpecificMesh` to be used with any `FiniteElement` that is based on the same geometric element type (see the section How to build a new geometric element for a more detailed discussion of geometric elements).
The minimum requirements for a specific mesh object are that it must:

- Construct the elements and the `Nodes`,
- Store pointers to the elements in the `Mesh::Element_pt` vector,
- Store pointers to the `Nodes` in the `Mesh::Node_pt` vector,
- Set the elements' pointers to their local `Nodes`
- Set the positions of the `Nodes`.

We suggest that all specific Mesh constructors also set up an auxiliary lookup scheme that stores information about the mesh boundaries. While this is not strictly required, it greatly facilitates the application of boundary conditions in complex domains. Firstly, the function Mesh::set_nboundary(...) must be called to specify the number of mesh boundaries. In addition, the generic Mesh class provides a function Mesh::add_↩ boundary_node(i,node_pt), which adds the Node pointed to by node_pt to the Mesh's i-th boundary and constructs the required look-up schemes, including a reverse lookup scheme that informs all Nodes which mesh boundaries (if any) they are located on. (In our simple 1D example there are only two boundaries, each consisting of a single Node.) Although this step is optional, it *is* required during mesh adaptation – if the reverse lookup scheme has not been set up, a warning is issued and the code execution stops when any mesh adaptation is attempted. **Implementation detail:** Nodes that are located on mesh boundaries must be defined as BoundaryNodes.

The majority of the work required to build a specific Mesh takes place in its constructor. Typically, the constructor creates the elements (of the type specified by the template parameter ELEMENT) and uses the elements' member function FiniteElement::construct_node(...) to create the new Nodes. The equivalent member function FiniteElement::construct_boundary_node(...) is used to create new Boundary↩ Nodes.
Here is the complete class definition for a simple, one-dimensional (line) mesh which discretises the 1D domain $x \in [0,1]$ using a specified number of equally-spaced elements.

```
template<class ELEMENT>
class OneDimMesh : public Mesh
{

public:

 /// Mesh Constructor. The argument is the desired number of elements
 OneDimMesh(const unsigned &n_element)
 {
  //Resize the vector of pointers to elements: there are n_element elements
  Element_pt.resize(n_element);

  //Construct the first element (Note the use of the template parameter)
  Element_pt[0] = new ELEMENT;

  //Construct the first node and add it to the Mesh::Node_pt vector
  //Note: The FiniteElement::construct_boundary_node(j) function
  //builds the element's j-th local node, and provides the functionality
  //that allows it to be located on a Mesh boundary -- essentially this
  //involves allocating additional storage to the Node.
  //The function obtains the Node's
  //characteristics (e.g. its spatial dimension, the number of
  //values to be stored, etc) from various virtual FiniteElement
  //member functions, such as FiniteElement::required_nvalue().
  //FiniteElement::construct_boundary_node(...) also
  //stores a pointer to the newly created Node in the element's own
  //Node_pt vector.
  //Finally, the function returns the pointer to the
  //newly created Node, so that it can be stored in the Mesh's Node_pt
  //vector, as done here:
  Node_pt.push_back(finite_element_pt(0)->construct_boundary_node(0));

  //Find the number of nodes per element (N.B. all elements are identical
  //so we can determine this value once and for all).
  unsigned n_node = finite_element_pt(0)->nnode();

  //Loop over the remaning nodes of the first element
  for(unsigned n=1;n<n_node;n++)
   {
    //Construct the next node and add it to the Mesh::Node_pt vector
    //Note that these interior nodes need not (and should not)
    //be boundary nodes, so they are created using the construct_node
    //function, which has the same interface as
    //construct_boundary_node()
    Node_pt.push_back(finite_element_pt(0)->construct_node(n));
   }

  //Loop over the remaining elements apart from the last
  for(unsigned e=1;e<(n_element-1);e++)
   {
    //Construct the e-th element
    Element_pt[e] = new ELEMENT;

    //The first local node of the e-th element is the last local node
    //of the (e-1)-th element. We MUST NOT construct the node twice.
    //Instead, we set the pointer in the e-th element to point to the
    //previously created node in the (e-1)-th element.
    finite_element_pt(e)->node_pt(0) =
```

```
      finite_element_pt(e-1)->node_pt(n_node-1);

    //Loop over the remaining nodes of the e-th element
    for(unsigned n=1;n<n_node;n++)
     {
      //Construct the next node and add it to the Mesh::Node_pt vector
      //Note that these interior nodes need not (and should not)
      //be boundary nodes, so they are created using the construct_node
      //function, which has the same interface as
      //construct_boundary_node()
      Node_pt.push_back(finite_element_pt(e)->construct_node(n));
     }
   } //End of loop over elements


  //Construct the final element
  Element_pt[n_element-1] = new ELEMENT;

  //The first local node of the final element is the last local node
  //of the penultimate element. We MUST NOT construct the node twice.
  //Instead, we set the pointer in the final element to point to the
  //previously created node in the penultimate element.
  finite_element_pt(n_element-1)->node_pt(0) =
   finite_element_pt(n_element-2)->node_pt(n_node-1);

  //Loop over the remaining central nodes of the final element
  for(unsigned n=1;n<(n_node-1);n++)
   {
    //Construct the next node and add it to the Mesh::Node_pt vector
    //Note that these interior nodes need not (and should not)
    //be boundary nodes, so they are created using the construct_node
    //function()
    Node_pt.push_back(finite_element_pt(n_element-1)->construct_node(n));
   }

  //Construct the final node and add it to the Mesh::Node_pt vector.
  //This node will be located on a boundary, and hence we use
  //the construct_boundary_node function.
  Node_pt.push_back(finite_element_pt(n_element-1)
                    ->construct_boundary_node(n_node-1));

  //We've now created all the nodes -- let's set their positions:

  //Find the total number of nodes
  unsigned n_global_node = nnode();

  //Loop over all nodes
  for(unsigned n=0;n<n_global_node;n++)
   {
    //Set the position of the node (equally spaced through the unit interval)
    Node_pt[n]->x(0) = double(n)/double(n_global_node-1);
   }
  //Set the boundary data:

  //There are two boundaries in this mesh
  set_nboundary(2);

  //Boundary 0 contains the first node in the mesh:
  add_boundary_node(0,Node_pt[0]);

  //Boundary 1 contains the final node in the mesh:
  add_boundary_node(1,Node_pt[n_global_node-1]);

 } // End of constructor

}; // End of OneDimMesh class.
```

To build a `OneDimMesh` with ten elements of type `SomeElement`, say, the `Problem` constructor would contain the following

```
Problem::mesh_pt() = new OneDimMesh<SomeElement>(10);
```

1.3.2 How to adapt an existing Mesh to a different domain shape

Given that mesh generation tends to be a fairly tedious process, one should always check whether it is possible to deform an existing `Mesh` into a shape that matches the required domain. Provided that the new domain has the same topology as the domain represented by the original `Mesh`, this can always be done by re-positioning the Nodes. Such "deformed" `Meshes` should be implemented via inheritance, by deriving the new `Mesh` from an existing one.

Here is an example that illustrates the procedure. Assume we wish to solve an equation in the 2D annular domain bounded (in polar coordinates) by $r \in [r_{min}, r_{max}]$ and $\varphi \in [\varphi_{min}, \varphi_{max}]$. Inspection of the list of available meshes shows that `oomph-lib` does not provide a mesh for this geometry. However, there is a

Mesh object, SimpleRectangularMesh, which provides a uniform discretisation of a 2D rectangular domain $x \in [0, L_x]$ and $y \in [0, L_y]$ with $N_x \times N_y$ quadrilateral elements. Since the topology of the two domains is identical, the annular mesh can be implemented in a few lines of code:

```
//======================================================================
/// AnnularQuadMesh, derived from SimpleRectangularQuadMesh.
//======================================================================
template<class ELEMENT>
class AnnularQuadMesh : public SimpleRectangularQuadMesh<ELEMENT>
{
  public:

 /// \short Constructor for angular mesh with n_r x n_phi
 /// 2D quad elements. Calls constructor for the underlying
 /// SimpleRectangularQuadMesh; then deforms the mesh so that it fits
 /// into the annular region bounded by the radii r_min and r_max
 /// and angles (in degree) of phi_min and phi_max.
 AnnularQuadMesh(const unsigned& n_r, const unsigned& n_phi,
                 const double& r_min, const double& r_max,
                 const double& phi_min, const double& phi_max) :
  SimpleRectangularQuadMesh<ELEMENT>(n_r,n_phi,1.0,1.0)
  {

   // The constructor for the  SimpleRectangularQuadMesh has
   // built the mesh with n_x x n_y = n_r x n_phi elements in the unit
   // square. Let's reposition the nodal points so that the mesh
   // gets mapped into the required annular region:

   // Find out how many nodes there are
   unsigned n_node=nnode();

   // Calculate the value of pi
   const double pi = 4.0*atan(1.0);

   // Loop over all nodes
   for (unsigned n=0;n<n_node;n++)
    {
     // Pointer to node:
     Node* nod_pt=node_pt(n);

     // Get the x/y coordinates
     double x_old=nod_pt->x(0);
     double y_old=nod_pt->x(1);

     // Map from the old x/y to the new r/phi:
     double r=r_min+(r_max-r_min)*x_old;
     double phi=(phi_min+(phi_max-phi_min)*y_old)*pi/180.0;

     // Set new nodal coordinates
     nod_pt->x(0)=r*cos(phi);
     nod_pt->x(1)=r*sin(phi);
    }
  }

};
```

1.3.3 Exercise:

Modify the OneDimMesh object so that it provides a piecewise uniform discretisation of the domain $x \in [0, 1]$: N_0 equally spaced elements are to be placed in the the region $x \in [0, \hat{x}]$, while N_1 elements are to be placed in the the region $x \in [\hat{x}, 1]$. Pass the parameters N_0, N_1 and $\hat{x}$ to the mesh constructor. Construct the modified mesh via inheritance from the basic OneDimMesh and include an error check to confirm that $0 < \hat{x} < 1$.

1.4 How to build a FiniteElement

oomph-lib provides fully-functional FiniteElements for the discretisation of a wide range of PDEs. Most of the existing elements are constructed in a three-level hierarchy, the base being the generic FiniteElement class. The next level in the hierarchy contains geometric elements (e.g. 1D line elements, 2D quad or triangular elements, 3D brick elements, etc.). These geometric classes form the bases for elements that implement the discretisation of particular PDEs. This hierarchy maximises the potential for code-reuse, because it allows many different specific elements to be derived from the same geometric element.
We shall discuss the implementation of new element types in two sections:

- How to implement a new system of equations as a specific FiniteElement
- How to build a new geometric element

1.4.1 How to implement a new system of equations as a specific FiniteElement

Assume that we wish to design a new element type, `SpecificElement`, say, that implements the discretisation of a system of PDEs on an existing geometric element. The `SpecificElement` class needs to implement the following functions:

- The function that specifies the number of nodal values required at each of the element's nodes:
  `FiniteElement::required_nvalue(n)`
- The function that computes the element's residual vector:
  `FiniteElement::get_residuals(residuals)`
- The function that computes the element's Jacobian matrix and its residual vector:
  `FiniteElement::get_jacobian(residuals,jacobian)`
- The output function:
  `FiniteElement::output(ostream)`

Most specific finite element classes will contain further member functions and member data to provide additional, problem-specific functionality. As a concrete example, we consider the `TwoNodePoissonElement`, a specific `FiniteElement` that provides an isoparametric discretisation of the 1D Poisson equation (1), based on the geometric element `TwoNodeGeometricElement`, to be discussed below:

```
class TwoNodePoissonElement : public TwoNodeGeometricElement
```

In addition to the `FiniteElement` member functions discussed above, this element provides a function that defines the source function $f(x)$,

```
  double f(const double &x) const
   {
    return double(Sign)*30.0*sin(sqrt(30.0)*x);
   }
```

where the sign of the source function is stored as private member data

```
  int Sign;
```

and can be set by the access function

```
  int& sign() {return Sign;}
```

We provide a member function that returns the (single) nodal value stored at a specified local node in the element,

```
  double u(const unsigned &n) {return node_pt(n)->value(0);}
```

Finally, it is good practice to implement a self-test function that provides a sanity check of all data before the problem is solved. `oomph-lib` already provides self-test functions for all fundamental objects. Additional tests can be added by overloading these. For instance, in our Poisson element, the `Sign` variable should only take the values ± 1. This can be tested with the following function, which also executes the `self_test()` function of the underlying `FiniteElement`:

```
  /// Self test function: The sign in the source function should
  /// only have the values +/- 1. Following the general oomph-lib convention,
  /// the self_test() returns 0 for success, and 1 for failure:
  unsigned self_test()
   {
    // Initialise success flag
    unsigned success=0;

    // Run the generic FiniteElement self test
    success=FiniteElement::self_test();

    // Do additional test for this function
    if ((Sign!=1)&&(Sign!=-1))
     {
      cout « "Sign of source function should be +/- 1," « std::endl;
      cout « "but it is:" « Sign « std::endl;
      success=1;
     }

    // Return success flag
    return success;

   } // End of self test
```

We will now discuss the implementation of the generic `FiniteElement` member functions for the specific `TwoNodePoissonElement`:

1.4.1.1 The function FiniteElement::required_nvalue(n)

The function returns the number of values that are stored at the element's `n-th` local node. In our scalar Poisson problem, each node stores one value:

```
  /// For the Poisson equation, only one value is stored at each node
  unsigned required_nvalue(const unsigned &n) const {return 1;}
```

The function is used by `FiniteElement::construct_node(...)` to determine the amount of storage to be allocated at each of the element's `Nodes`.

1.4.1.2 The function FiniteElement::get_residuals(residuals)

This function calculates the element's residual vector whose entries are labelled by the *local* equation numbers. (Consult the Introduction for a detailed discussion of the weak form of Poisson's equation and its finite element discretisation which produces the discrete residual implemented here.) The spatial integration over the element is performed by the (default) integration scheme specified in the underlying geometric element. Note that it should (must) not be assumed that the residuals vector has been set to zero before the function call.

```
void get_residuals(Vector<double> &residuals)
 {
  //Find the number of degrees of freedom (unpinned values) in the element
  unsigned n_dof = ndof();

  //Initialise all the residuals to zero
  for(unsigned i=0;i<n_dof;i++) {residuals[i] = 0.0;}

  //Find the number of nodes in the element
  unsigned n_node = nnode();

  //Allocate memory for shape functions and their derivatives:
  // There's one shape function for each node:
  Shape psi(n_node);

  // Each of the n_node shape functions has one derivative with
  // respect to the single local coordinate:
  DShape dpsidx(n_node,1);

  //Storage for the single local coordinate
  Vector<double> s(1);

  //Find the number of integration points in the underlying
  //geometric element's integration scheme
  unsigned n_intpt = integral_pt()->nweight();

  //Loop over the integration points
  for(unsigned ipt=0;ipt<n_intpt;ipt++)
   {
    //Set the value of the local coordinate to be the integration
    //scheme's knot point
    s[0] = integral_pt()->knot(ipt,0);

    //Find the weight of the integration scheme at this knot point
    double w = integral_pt()->weight(ipt);

    //Find the shape functions and their derivatives at the knot point.
    //This function is implemented in FiniteElement.
    //It also returns the Jacobian of the mapping from local to
    //global coordinates.
    double J = dshape_eulerian(s,psi,dpsidx);

    //Premultiply the weight and the Jacobian
    double W = w*J;

    //Allocate storage for the value of the field variable u,
    //its derivative and the global position at the knot point.
    //Initialise them all to zero.
    double interpolated_x=0.0, interpolated_u=0.0, interpolated_dudx=0.0;

    //Calculate the interpolated values by  looping over the shape
    //functions and summing the appropriate contributions
    for(unsigned n=0;n<n_node;n++)
     {
      interpolated_x += nodal_position(n,0)*psi[n];
      interpolated_u += u(n)*psi[n];
      interpolated_dudx += u(n)*dpsidx(n,0);
     }

    // Evaluate the source function
    double source=f(interpolated_x);

    //ASSEMBLE THE RESIDUALS

    //Loop over the test functions (same as the shape functions
    //since we're implementing an isoparametric element)
    for(unsigned l=0;l<n_node;l++)
     {
      //Get the local equation number
      //The variable is the first (only) value stored at the nodes
      int local_eqn_number = nodal_local_eqn(l,0);

      //If the equation is not a Dirichlet boundary condition
      if(local_eqn_number >= 0)
       {
        //Add body force/source term here
```

```
          residuals[local_eqn_number] += source*psi[l]*W;

          //Add the Poisson bit itself
          residuals[local_eqn_number] += interpolated_dudx*dpsidx(l,0)*W;
         }
       }

     } //End of loop over the integration points

   } //End of function
```

1.4.1.3 The function FiniteElement::get_jacobian(residuals,jacobian)

This function calculates the element's Jacobian matrix and its residual vector, performing the spatial integration with the (default) integration scheme defined in the underlying geometric element. The entries in the Jacobian matrix and the residual vector are labelled by the *local* equation numbers. Note that neither the residuals vector nor the Jacobian matrix should be assumed to have been initialised to zero before the function call.

```
  void get_jacobian(Vector<double> &residuals, DenseMatrix<double> &jacobian)
   {
    //First, calculate the residuals
    get_residuals(residuals);

    //Find the number of degrees of freedom (unpinned values) in the element
    unsigned n_dof = ndof();

    //Initialise all entries of the Jacobian matrix to zero
    for(unsigned i=0;i<n_dof;i++)
     {
      for(unsigned j=0;j<n_dof;j++) {jacobian(i,j) = 0.0;}
     }

    //Find the number of nodes in the element
    unsigned n_node = nnode();
    //Allocate memory for shape functions and their derivatives
    Shape psi(n_node);
    DShape dpsidx(n_node,1);

    //Storage for the local coordinate
    Vector<double> s(1);

    //Find the number of integration points in the underlying
    //geometric element's integration scheme
    unsigned n_intpt = integral_pt()->nweight();

    //Loop over the integration points
    for(unsigned ipt=0;ipt<n_intpt;ipt++)
     {
      //Set the value of the local coordinate to be the integration
      //scheme's knot point
      s[0] = integral_pt()->knot(ipt,0);

      //Find the weight of the integration scheme at this knot point
      double w = integral_pt()->weight(ipt);

      //Find the shape functions and their derivatives at the knot point.
      //This function is implemented in FiniteElement.
      //It also returns the Jacobian of the mapping from local to
      //global coordinates.
      double J = dshape_eulerian(s,psi,dpsidx);

      //Premultiply the weight and the Jacobian
      double W = w*J;

      //ASSEMBLE THE JACOBIAN TERMS

      //Loop over the test (shape) functions
      for(unsigned l=0;l<n_node;l++)
       {
        //Get the local equation number
        //The variable is the first (only) value stored at the nodes
        int local_eqn_number = nodal_local_eqn(l,0);

        //If the equation is not a Dirichlet boundary condition
        if(local_eqn_number >= 0)
         {
          //Loop over the degrees of freedom
          for(unsigned l2=0;l2<n_node;l2++)
           {
            //Get the local degree of freedom number
            //The variable is the first (only) value stored at the nodes
            int local_dof_number = nodal_local_eqn(l2,0);

            //If the degree of freedom is not pinned
            if(local_dof_number >= 0)
```

```
            {
             //Add the contribution to the Jacobian
             jacobian(local_eqn_number,local_dof_number) +=
              dpsidx(l2,0)*dpsidx(l,0)*W;
            }
          }
        }
      }
     } //End of loop over the integration points

   } //End of function
```

Note: There is a large amount of code duplication between the `get_residuals()` and `get_jacobian()` functions. To avoid this, we usually implement the computation of the residual vector and the Jacobian matrix in a single function containing the loop over the integration points. We then use a boolean flag as an additional argument to determine whether the Jacobian matrix should be assembled, e.g.

```
void get_generic_residual_contribution(Vector<double>& residuals,
                                       DenseMatrix<double>& jacobian,
                                       bool flag)
```

1.4.1.4 The function FiniteElement::output(out_stream)

The output function is used to print the nodal values over the element

```
  void output(ostream &output)
   {
    //Read out the number of nodes in the element
    unsigned n_node = nnode();

    //Loop over the nodes and print out the global coordinate
    //and value of the field variable, u, at each node
    for(unsigned n=0;n<n_node;n++)
     {
      output « nodal_position(n,0) « " " « u(n) « std::endl;
     }
   } //End of function
```

1.4.2 Exercises:

- Implement the function `get_generic_residual_contribution(...)` as a private member function of the `TwoNodePoissonElement`. Use the function to avoid the large amount of code duplication between `get_residuals(...)` and `get_jacobian(...)` by re-writing these functions as follows:

  ```
  /// \short Calculate the elemental contributions to the residuals for
  /// the weak form of the Poisson equation
  void get_residuals(Vector<double> &residuals)
   {
    // Set flag for not computing the Jacobian
    bool flag=false;

    // Dummy Jacobian
    DenseMatrix<double> jacobian;

    // Compute the residuals only:
    get_generic_residual_contribution(residuals,jacobian,flag);
   }
  ```

  and

  ```
  /// \short Calculate the elemental contribution to the Jacobian
  /// matrix dR_{i}/du_{j} used in the Newton method
  void get_jacobian(Vector<double> &residuals, DenseMatrix<double> &jacobian)
   {
    // Set flag for computing the Jacobian matrix
    bool flag=true;

    // Compute the residuals and the Jacobian matrix:
    get_generic_residual_contribution(residuals,jacobian,flag);
   }
  ```

- The implementation of the specific source function $f(x)$ in `TwoNodePoissonElement::f(...)` makes it impossible to use the `TwoNodePoisson↩Element` to solve the Poisson equation with any other source functions, even though the discretisation of the ODE would otherwise be completely identical. A better implementation would allow the "user" to specify a different source function without having to change the implementation of the element class itself.

  Here is a suggestion for an improved implementation, which illustrates the procedure employed in the existing elements in `oomph-lib`:

  - Introduce a public typedef into the element class to define the required format of the function pointer:

    ```
    /// \short Function pointer to source function:
    ```

```
/// The source function returns the value of the
/// source function at the global coordinate x.
typedef double (*PoissonSourceFctPt)(const double& x);
```

– Add a function pointer to the private data of the `TwoNodePoissonElement` class and initialise it to NULL in the constructor:

```
/// \short Function pointer to source function (initialised to
/// NULL in the constructor)
PoissonSourceFctPt Source_fct_pt;
```

– Provide an access function for the source function pointer:

```
/// \short Access function to pointer to source function
PoissonSourceFctPt& source_fct_pt() {return Source_fct_pt;}
```

– Re-write the `TwoNodePoissonElement::f(...)` function, so that it evaluates the (global) function pointed to by the source function pointer. If possible, provide a default value for the case when the function pointer has not been set:

```
/// Evaluate source function at Eulerian position x
double f(const double& x) const
 {
  //If no source function has been set, return zero
  //so that the Poisson equation defaults to a Laplace equation.
  double source=0.0;
  if(Source_fct_pt!=0)
   {
    // Evaluate source function
    source = (*Source_fct_pt)(x);
   }
  return source;
 }
```

The pointer to the source function (typically defined in a suitable namespace in the "user's" driver code) can now be set in the Problem constructor.

- Generalise the `TwoNodePoissonElement` to a `TwoNodeSelfAdjointElement` that implements the isoparametric discretisation of the self-adjoint ODE

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(a(x)\frac{\mathrm{d}u}{\mathrm{d}x}\right)+b(x)u(x)=f(x). \qquad (3)$$

Use function pointers to allow the "user" to specify the coefficient functions $a(x)$, $b(x)$ and $f(x)$.

1.4.3 How to build a new geometric element

Geometric elements define the geometry of a `FiniteElement`. They are usually implemented as distinct classes that can then be used to create a number of `FiniteElements` each discretising a specific PDE, but with the same underlying geometrical representation. For this purpose, each geometric `FiniteElement` must implement the following functions:

- The constructor
  `SomeGeometricFiniteElement::SomeGeometricFiniteElement()`

  must perform the following actions:

  – Set the number of Nodes by calling the protected member function
  `FiniteElement::set_n_node(n_node).`

  – Set the dimension of the element by calling the protected member function
  `FiniteElement::set_dimension(dim)`

  Here the dimension of the element is defined as the number of local coordinates required to parametrise its shape. (Note that this is not necessarily the same as the spatial dimension of the problem – we can have one-dimensional `FiniteElements` in 3D space, say.)

  – **Optional:** Set the spatial integration scheme by calling the protected member function
  `FiniteElement::set_integration_scheme(&integration_scheme)`

  – **Optional:** If the spatial dimension (= the number of Eulerian coordinates) required at all Nodes in the element is different from the spatial dimension of the element then set the "nodal dimension" by calling the protected member function
  `FiniteElement::set_nodal_dimension(dim)`

  For example in shell elements, a two-dimensional surface is embedded in a three-dimensional space, so the elemental dimension is two, but the nodal dimension is three.

  - **Optional:** If the Eulerian coordinate (position) is interpolated using generalised coordinates, e.g. vector-valued shape functions, or Hermite-type interpolation, set the number of generalised coordinates or "position types" by using the protected member function
    `FiniteElement::set_n_nodal_position_type(n_position_type)`

- The function that computes the element's (geometric) shape functions at specified values of the local coordinates. These shape functions are primarily used to implement the mapping between the element's local coordinates and the global (Eulerian) coordinates.
  The geometric shape functions *can* be (and often are) also used in specific `FiniteElements` to interpolate the unknown function(s) between the nodal values.
  `FiniteElement::shape(s,psi)`

It is usually necessary to implement the following additional functions:

- The function that computes the derivatives of the element's shape functions with respect to the local coordinates,
  `FiniteElement::dshape_local(s,psi,dpsids)`

- The function that specifies the number of Nodes along the element's 1D "edges"

  `FiniteElement::nnode_1d()`

Further member functions and member data may be defined as and when required. For instance, all specific `FiniteElements` must store a pointer to an (instantiated) spatial integration scheme. It is good practice to provide a default integration scheme for each geometric element and to ensure its instantiation by making it a static data member of the geometric element class. This allows the constructor of the geometric element to set the pointer to the default integration scheme. If the default is not appropriate for a specific derived `FiniteElement`, the default assignment can be over-written in the constructor of the derived class.
As a concrete example, we consider the implementation of the one-dimensional, two-node geometric element, `TwoNodeGeometricElement`, that uses linear shape functions to define the mapping between the element's local and global coordinates. The element is derived from the `FiniteElement` base class,

```
class TwoNodeGeometricElement : public FiniteElement
```

and it uses a one-dimensional, two-point Gauss rule as the default spatial integration scheme

```
 /// Integration scheme that will be used to integrate over the element.
 /// Simple Gaussian quadrature in one dimension, with two Gauss points.
 static Gauss<1,2> Default_spatial_integration_scheme;
```

Here is the implementation of the generic `FiniteElement` member functions for our specific geometric `FiniteElement:`

1.4.3.1 The constructor: TwoNodeGeometricElement()

The constructor sets the number of local nodes in the element (2), sets the dimension of the element (1) and sets the pointer to the spatial integration scheme to be the default that is defined and instantiated as a static data member of the class:

```
 TwoNodeGeometricElement()
  {
   //Linear interpolation requires two Nodes per element.
   //In fact, calling this function merely provides storage for
   //the pointers to the Nodes and initialises the pointers to NULL.
   //The Nodes themselves are created during the mesh generation
   //process by the functions FiniteElement::construct_node(...)
   //which stores the pointers to the newly created Nodes
   //in the element's own internal storage.
   this->set_n_node(2);

   //The element is one-dimensional
   this->set_dimension(1);

   //Set the pointer to the spatial integration scheme
   set_integration_scheme(&Default_spatial_integration_scheme);
  }
```

1.4.3.2 The function FiniteElement::shape(s,psi)

This function returns the shape functions at the local element coordinate, s. The shape functions are used to interpolate the position, and also, any data values stored at nodes in derived, isoparametric elements.

```
 void shape(const Vector<double> &s, Shape &psi) const
  {
   //There are two shape functions (one per node)
   //In terms of the local coordinate, s[0]:
```

```
 //Node 0 is at s[0] = -1.0
 //Node 1 is at s[0] =  1.0

 //psi[0] takes the value one at node 0 and zero at node 1
 psi[0] = 0.5*(1.0 - s[0]);

 //psi[1] takes the value one at node 1 and zero at node 0
 psi[1] = 0.5*(1.0 + s[0]);
}
```

1.4.3.3 The function FiniteElement::dshape_local(s,psi,dpsids)

This function returns the shape functions and their derivatives with respect to the local coordinates

```
void dshape_local(const Vector<double> &s, Shape &psi, DShape &dpsids) const
 {
  //Call the shape functions
  shape(s,psi);

  //The derivative of psi[0] wrt s[0] is -0.5
  dpsids(0,0) = -0.5;

  //The derivative of psi[0] wrt s[0] is 0.5
  dpsids(1,0) = 0.5;
 }
```

1.4.3.4 The function FiniteElement::nnode_1d()

Return the number of nodes along the element's 1d "edge" – for a one-dimensional element, this is obviously the same as the number of nodes:

```
unsigned nnode_1d() const {return 2;}
```

1.4.4 Exercises:

- Change the `TwoNodeGeometricElement` to a `ThreeNodeGeometricElement` in which quadratic interpolation is used to interpolate the Eulerian coordinates between the nodal points. Use this element as a basis for a `ThreeNodePoissonElement` and convince yourself that changing the `TwoNode↩ PoissonElement` to a `ThreeNodePoissonElement` only requires the change of a single line of code!

- Use templating to generalise the geometric elements to an arbitrary number of nodes so that the class

  ```
  template <unsigned NNODE>
  class GeometricLineElement : public FiniteElement
  ```

  represents 1D line elements with `NNODE` nodes. Consider carefully which member functions you can implement in generality and which member functions require specialised implementations. Provide the specialised member functions for elements with two, three and four nodes.

- Use the templated `GeometricLineElements` to implement an equivalent generalisation of the `Two↩ NodePoissonElement` and `ThreeNodePoissonElement` classes to the general class

  ```
  template <unsigned NNODE>
  class PoissonLineElement : public GeometricLineElement<NNODE>
  ```

- Validate all elements against the exact solution and confirm the theoretical error estimate which predicts that if the domain $x \in [0,1]$ is discretised with N equally spaced n-node line elements, we have

$$e = \sqrt{\int_0^1 \left(u_{FE}(x) - u_{exact}(x)\right)^2 dx} \sim h^{(n-1)} \quad \text{as } N \to \infty$$

  where $h = 1/N$.

1.5 Further comments

We reiterate that the main purpose of this document is to provide a *quick* introduction to `oomph-lib's` fundamental objects. The exercises have already highlighted several undesirable features of the simple example classes which could easily be improved to facilitate the (re-)use of the classes in different problems. Here we shall briefly discuss some further modifications that we regard as good practice, and which tend to be implemented in the existing classes in `oomph-lib`:

- Dimension-independent implementation
- Using pre-computed shape functions and the StorableShapeElement<ELEMENT> class
- Further sub-division of specific element classes – equation classes.
- Implement extensive (but optional) self tests in all classes
- The use of "magic numbers"
- Facilitating the use of specific elements in multi-physics problems

1.5.1 Dimension-independent implementation

The exercises in the section How to build a new geometric element suggest the use of templating to implement families of objects. This idea also permits the implementation of (geometric and specific) `FiniteElements` in arbitrary spatial dimensions. For instance, `oomph-lib's QElement` class represents the family of (line/quadrilateral/brick-shaped) geometric elements with an arbitrary number of nodes along the elements' 1D edges. Thus, a `QElement<1,3>` is a three-node 1D line element, a `QElement<3,2>` is an eight-node 3D brick element, etc.
These geometric elements naturally form the basis for corresponding specific elements, such as the `QPoissonElement` family which provides an isoparametric discretisation of the Poisson equation (in an arbitrary spatial dimension), based on the `QElements` with the same template parameters.

1.5.2 Using pre-computed shape functions and the StorableShapeElement<ELEMENT> class

The computation of the element's residual vector and the Jacobian matrix requires the evaluation of the element's shape functions, ψ_j, and their derivatives with respect to the global coordinate, $\mathrm{d}\psi_j/\mathrm{d}x$, at the element's integration points. In our simple example class, `TwoNodePoissonElement`, we (re-)compute these functions "on the fly", using the function
`FiniteElement::dshape_eulerian(s,psi,dpsidx)`
where the local coordinate of the integration point is passed in the (one-dimensional) vector `s`.
The re-computation is wasteful because:

- The values of the shape functions and their derivatives, evaluated at the integration point, are independent of the value of the `Sign` variable. Their re-computation during the second solve could therefore be avoided if we stored their values in a suitable container.
- The values of the shape functions at the integration points are identical for all elements. Therefore we only need to provide storage for the shape function values in a single, representative element. The other elements can access the values via pointers.

To circumvent these potential inefficiencies, `oomph-lib` provides alternative interfaces for various functions that compute shape functions and their derivatives:

- Calls to the function
  `FiniteElement::shape(s,psi)`

  which computes the shape function at a given local coordinate, `s`, can be replaced by
  `FiniteElement::shape_at_knot(int_point,psi)`

  where the `unsigned` argument `int_point` identifies the integration point (as specified by the element's spatial integration scheme). In "standard" `FiniteElements` the function `FiniteElement::shape_at_knot(...)` simply determines the position of the integration point and calls `FiniteElement::shape(...)` and so the shape functions are still computed "on the fly". The templated class `StorableShapeElement<ELEMENT>`, however, may be used to upgrade any class derived from `FiniteElement` into a class that **can** store the values of the shape functions and their derivatives at the integration points. The function `shape_at_knot(...)` is overloaded so that when called for the first time, it computes the values of the shape function at all integration points and stores them for future reference. In subsequent calls, the function returns the stored values, rather than re-computing them.

- There are equivalent alternatives for the functions that compute the shape functions and their derivatives with respect to the local coordinates:
  `FiniteElement::dshape(s,psi,dpsids)`

  can be replaced by
  `FiniteElement::dshape_at_knot(int_point,psi,dpsids)`

  while
  `FiniteElement::d2shape(s,psi,dpsids,d2psids)`

  can be replaced by
  `FiniteElement::d2shape_at_knot(int_point,psi,dpsids,d2psids)`

These functions are all overloaded in the `StorableShapeElement<ELEMENT>` class and by default the overloaded functions store the pre-computed values of the shape functions and their derivatives locally within each element. This implementation ensures data locality and should increase the speed of access to the stored values. However, it can also create significant storage overheads. `oomph-lib` therefore provides the function
`StorableShapeElement<ELEMENT>::set_shape_local_stored_from_element(...)`
which frees up the storage in the element and replaces it by pointer-based access to the values stored in another element. Typically, all elements in a mesh are of the same type and use the same spatial integration scheme and so the values of the shape functions need only be stored in one element.
Derivatives of the element's shape functions with respect to the global Eulerian coordinates are generally computed by
`FiniteElement::dshape_eulerian(s,psi,dpsidx)`
and
`FiniteElement::d2shape_eulerian(s,psi,dpsidx,d2psidx)`
These functions also return the Jacobian of the mapping between the local and global coordinates – the Jacobian can also be computed independently from
`FiniteElement::J_eulerian(s)`
The derivatives of the shape functions with respect the global coordinates depend on the nodal positions, and it is not safe to assume that they will remain constant. For instance in moving boundary problems, the position of the nodes is determined as part of the solution. By default, the functions
`StorableShapeElement<ELEMENT>::dshape_eulerian_at_knot(int_point,psi,dpsidx)`
`StorableShapeElement<ELEMENT>::d2shape_eulerian_at_knot(int_point,psi,dpsidx,d2psidx)`
and
`StorableShapeElement<ELEMENT>::J_eulerian_at_knot(int_point)`
re-compute the derivatives at the specified integration point "on the fly". If the "user" is confident that in his/her problem, the nodal positions will not change, the values can be pre-computed by calling
`StorableShapeElement<ELEMENT>::pre_compute_dshape_eulerian_at_knots()`
`StorableShapeElement<ELEMENT>::pre_compute_d2shape_eulerian_at_knots()`
`StorableShapeElement<ELEMENT>::pre_compute_J_eulerian_at_knots()`
Once these functions have been called, any subsequent calls to the `*_eulerian_at_knot(...)` functions return the stored values. To revert to the case in which the derivatives are re-computed "on the fly", the storage for the derivatives must be deleted by
calling
`StorableShapeElement<ELEMENT>::delete_dshape_eulerian_stored()`
Notes:

- In the (unlikely!) case that an element's spatial integration scheme is changed during the code execution, all stored values are automatically re-computed.
- The class `StorableShapeSolidElement<ELEMENT>` provides equivalent overloaded functions for derivatives with respect to the element's Lagrangian coordinates that are used in solid mechanics problems.

1.5.3 Further sub-division of specific element classes – equation classes.

We suggested that any newly developed elements should be constructed in a three-level hierarchy, `Finite↩ Element` – Geometric Element – Specific `FiniteElement`. Most finite elements in `oomph-lib` incorporate an additional intermediate "equation class" which implements the computation of the element residual vector and the element Jacobian matrix in terms of abstract shape and test functions, defined as pure virtual functions in the "equation class". This makes it easy to change the specific element formulation, without having to re-implement the weak form of the governing equation.
Note that different element types may store the same physical variable at different locations. For example, the pressure in the Navier–Stokes equations may be stored as internal `Data` (discontinuous) or nodal `Data` (continuous). Particular equation classes may require internal numbering schemes that store the appropriate local equation numbers for each physical variable. These schemes must be assembled for each specific element in the function

`GeneralisedElement::assign_additional_local_eqn_numbers()`, which is called from within `Problem::assign_eqn_numbers()`.
As an example, consider the weak form of the 2D advection diffusion equation

$$\int \left(\mathsf{Pe} \sum_{i=1}^{2} w_i \frac{\partial u}{\partial x_i} \psi_l^{(test)} + \sum_{i=1}^{2} \frac{\partial u}{\partial x_i} \frac{\partial \psi_l^{(test)}}{\partial x_i} \right) dA = 0,$$

where the Peclet number, Pe , and the "wind" w_i $(i = 1, 2)$ are given. We expand the unknown function $u(x_1, x_2)$ in terms of the (global) basis functions $\psi_k^{(basis)}(x_1, x_2)$, $(k = 1, ..., N)$,

$$u(x_1, x_2) = \sum_{k=1}^{N} U_k \, \psi_k^{(basis)}(x_1, x_2),$$

where N is the total number of nodes in the mesh. The mapping between the element's local and global coordinates is represented in terms of the local shape functions $\psi_j^{(shape)}(s_1, s_2)$ as

$$x_i = \sum_{j=1}^{n} X_{ij} \, \psi_j^{(shape)}(s_1, s_2)$$

where n is the number of nodes in the element.
The following sketch illustrates how this discretisation is implemented in `oomph-lib's QAdvectionDiffusionElement` – an isoparametric, quadrilateral element, based on the Galerkin discretisation of the weak form with $\psi_j^{(shape)} = \psi_j^{(test)} = \psi_j^{(basis)}$

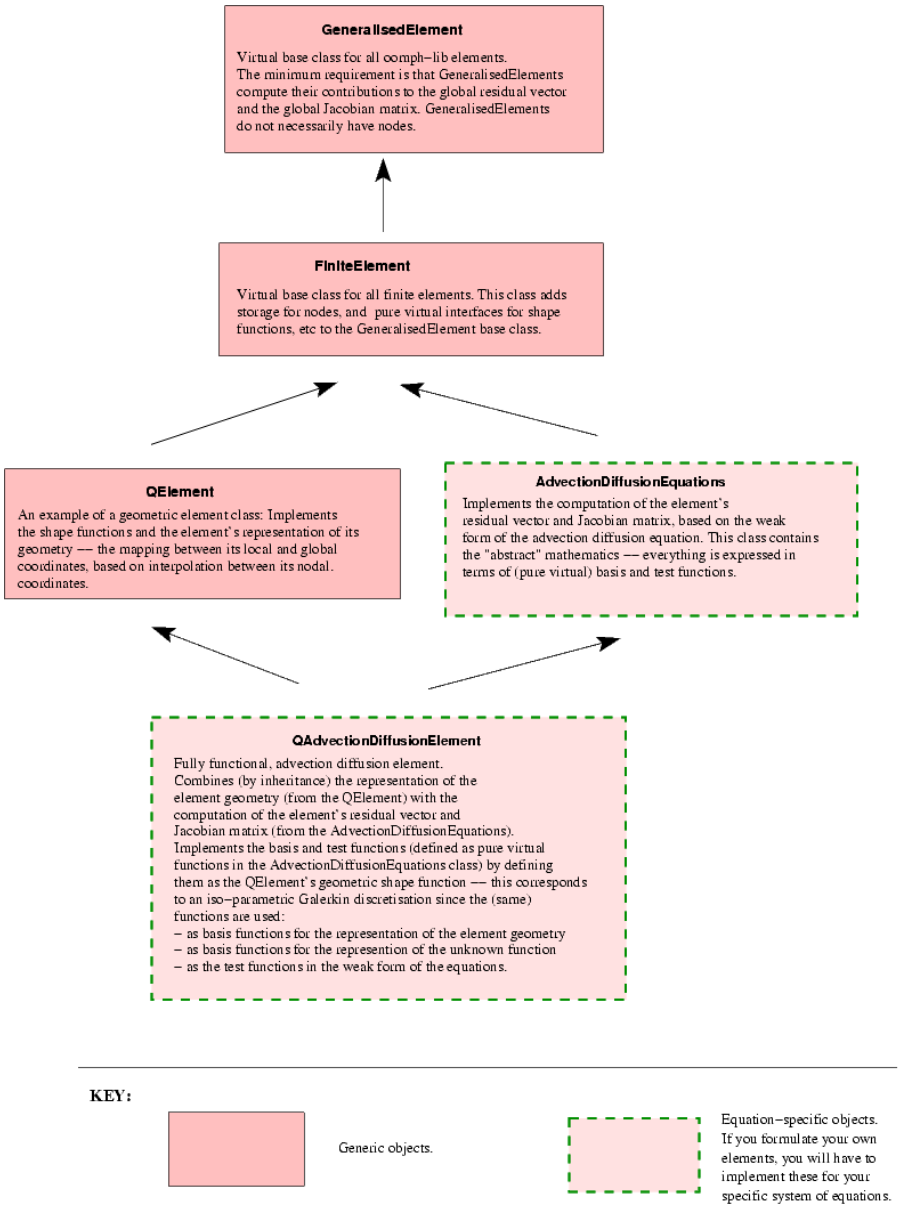


Figure 1.1 Typical inheritance diagram for oomph-lib elements

At large Peclet number, the Galerkin discretisation of the advection diffusion equation is well-known to produce spurious "wiggles" in the solution. These can be suppressed by SUPG stabilisation which employs test functions, $\psi_j^{(test)}$, that differ from the basis function, $\psi_j^{(basis)}$,

$$\psi_j^{(test)} = \psi_j^{(basis)} + \tau^{(SUPG)} \sum_{i=1}^{2} w_i \frac{\partial \psi_j^{(basis)}}{\partial x_i},$$

where $\tau^{(SUPG)}$ is a stabilisation parameter. This can be implemented with a trivial change to the `QAdvection↩DiffusionElement` class – the `QSUPGAdvectionDiffusionElement` simply provides a different implementation of the test functions.

1.5.4 Implement extensive (but optional) self tests in all classes

We have already illustrated the use of `oomph-lib's` generic `self_test()` functions. The top-level `Problem::self_test()` function performs a systematic test of all fundamental objects involved in a specific problem and can be used to diagnose any problems. It is good practise to implement further `self↩_test()` functions in any newly developed classes. The generic `self_test()` functions are defined to be virtual functions in `oomph-lib's` fundamental objects and can be overloaded. Obviously, the `self↩`

_test() function in a specific derived object should still call the self_test() function of the underlying fundamental object. The TwoNodePoissonElement::self_test() function, listed in in the section How to implement a new system of equations as a specific FiniteElement illustrates the procedure.
While frequent sanity checks are helpful during code-development, they can introduce significant overheads into the code execution. oomph-lib therefore provides a compiler flag PARANOID, which allows the execution of sanity checks to be switched on or off. When developing new classes, sanity checks should be implemented to catch any potential problems, but the relevant code should be surrounded by ifdef/endif statements to allow the tests to be disabled. Here is an example:

```
#include <typeinfo>

 [...]

 // Recast to a different pointer type
 SomeOtherObject* other_pt=dynamic_cast<SomeOtherObject*>(some_pt);

#ifdef PARANOID
 if (other_pt==0)
  {
  std::ostringstream error_stream;
    error_stream << "Failed to cast some_pt to SomeOtherObject* "
                 << std::endl;
    error_stream << "The pointer some_pt points to an object of type: " <<
                 << typeid(some_pt).name() << std::endl;
    throw OomphLibError(error_stream.str(),
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
   }
#endif

[...]
```

Note that we throw a specific OomphLibError object after catching the error – this allows the provision of more explicit (and hopefully more meaningful) error messages.
Many access functions that provide indexed access to a private container, do, in fact, access a private STL vector. Explicit range checking for these (frequent!) cases can be avoided by changing the container to oomph-lib's Vector class. The Vector class performs automatic range checking, if oomph-lib's generic library is compiled with the RANGE_CHECKING flag set (i.e. if -DRANGE_CHECKING is specified as a compilation flag for the C++ compiler). For access functions that do not use the Vector class you should implement your own range checks using the RANGE_CHECKING compiler flag.

1.5.5 The use of "magic numbers"

In general, we set the equation number to -1 to indicate that a nodal value is pinned, i.e. prescribed by boundary conditions. This convention is used throughout oomph-lib, but the direct use of -1 as a (bad!) "magic number" is generally avoided. Instead we refer to the static data member of the Data class

```
static long Data::Is_pinned
```

which is (of course) set to -1. Similarly, when nodes are created, the equation numbers of their value(s) are initialised to a value of -10, represented by the static data member

```
static int Data::Is_unclassified
```

This allows the function Problem::self_test() to check if any values have not been classified as pinned or free.

1.5.6 Facilitating the use of specific elements in multi-physics problems

As the name of the library suggests, oomph-lib's existing "single-physics" elements are (and any newly designed ones should be) designed so that they can easily be used in multi-physics problems. We anticipate two types of multi-physics interactions:

- The solution of one system of equations affects the geometry of the domain in which another equation is solved – this occurs, for instance, in fluid-structure interaction problems.
- Two physical phenomena occur in the same domain and interact with each other, so that forcing terms in one equation depend on the solution of another – this occurs, for instance, in thermo-elasticity problems where the temperature distribution (determined by the solution of an unsteady heat equation, say) affects the "growth term" in the equations of elasticity.

Multi-physics elements for such problems are usually implemented by combining two (or more) single-physics elements via multiple inheritance. The combined, multi-physics element will obviously have a larger number of degrees of freedom than the constituent single-physics elements. The combined element typically calls the get↩_residuals(...) function of the two constituent elements and concatenates their residual vectors. When

computing the Jacobian matrix, the "single physics" elements provide the diagonal blocks for the Jacobian matrix of the multi-physics element, while the off-diagonal interaction blocks must be computed separately. The details of the implementation vary from problem to problem. However, any single-physics element must satisfy the following requirements if it is to be used as a base class for derived multi-physics elements:

- Rule 1: Elements must only initialise their own entries in the element residual vector and the element Jacobian matrix
- Rule 2: Forcing functions, etc. should be implemented as uniquely-named virtual functions
- Rule 3: Nodal values should only be accessed indirectly via index functions that can be overloaded in derived multi-physics ele
- Rule 4: Time derivatives should be implemented in the ALE formulation to allow for moving meshes.

1.5.6.1 Rule 1: Elements must only initialise their own entries in the element residual vector and the element Jacobian matrix

The computation of the element residual vector and the element Jacobian matrix typically involves a loop over the element's integration points, in which each integration point adds a further contribution to the entries. In multi-physics problems it is crucial that the initialisation of the residual vector and the Jacobian matrix only sets the element's "own" entries to zero.

To illustrate this point, consider what would happen if we used the `TwoNodePoissonElement::get_jacobian(...)` function, discussed in the section The function FiniteElement::get_jacobian(residuals,jacobian), in a derived multi-physics element, which combines the `TwoNodePoissonElement` with another element, `TwoNodeSomeOtherEquationElement`, say. Assume that we implement the function `get_jacobian(...)` of the combined element so that it first calls the function `TwoNodeSomeOtherEquationElement::get_jacobian(...)` to determine the first diagonal block in the combined Jacobian matrix. When we call `TwoNodePoissonElement::get_jacobian(...)` to compute the entries in the second diagonal block, the initialisation loop

```
//Find the number of degrees of freedom (unpinned values) in the element
 unsigned n_dof = ndof();
//Initialise all entries of the Jacobian matrix to zero
for(unsigned i=0;i<n_dof;i++)
 {
  for(unsigned j=0;j<n_dof;j++) {jacobian(i,j) = 0.0;}
 }
```

would initialise the *entire* Jacobian matrix, thus wiping out the entries that were already computed by `TwoNodeSomeOtherEquationElement::get_jacobian(...)`.

The strategy used in `oomph-lib` to permit the easy combination of elements is to use the two protected member functions of the `GeneralisedElement` class:

```
virtual void GeneralisedElement::fill_in_contribution_to_residuals(Vector<double>&residuals);
```

and

```
virtual void GeneralisedElement::fill_in_contribution_to_jacobian(Vector<double> &residuals,
                                                                  DenseMatrix<double> &jacobian);
```

These functions DO NOT initialise the entries of the residuals vector or the Jacobian matrix. Instead, the functions merely add the appropriate contributions to the vector and the matrix entries. The default version of the `get_residuals()` and `get_jacobian()` functions, defined in `GeneralisedElement`, are simple wrappers that initialise the residuals and Jacobian to zero and then call the appropriate `fill_in_contribution...` function.

```
virtual void GeneralisedElement::get_residuals(Vector<double> &residuals)
{
 //Zero the residuals vector
 residuals.initialise(0.0);

 //Add the elemental contribution to the residuals vector
 fill_in_contribution_to_residuals(residuals);
}

virtual void GeneralisedElement::get_jacobian(Vector<double> &residuals,
                                              DenseMatrix<double> &jacobian)
{
 //Zero the residuals vector
 residuals.initialise(0.0);

 //Zero the Jacobian matrix
 jacobian.initialise(0.0);

 //Add the elemental contribution to the residuals vector and Jacobian
 fill_in_contribution_to_jacobian(residuals,jacobian);
}
```

The `get_residuals` function, for example, can thus be overloaded in a multi-physics element, as follows:

```
/// Multi-physics element, created by multiple inheritance
class SomeMultiPhysicsElement : public virtual TwoNodePoissonElement,
                                public virtual TwoNodeSomeOtherEquationElement
{

 [...]

 /// Residual vector of the combined element is made from the entries
 /// of the constituent single-physics elements
 virtual void get_residuals(Vector<double> &residuals)
  {
   //Zero the residuals vector
   residuals.initialise(0.0);

  //Add the first elemental contribution to the residuals vector
  TwoNodePoissonElement::fill_in_contribution_to_residuals(residuals);

   //Add the second elemental contribution to the residuals vector
   TwoNodeSomeOtherEquationElement::fill_in_contribution_to_residuals(residuals);
  }


  [...]

};
```

It is, therefore, recommended that authors of "single-physics" elements, overload `fill_in_contribution_to_residuals(...)` and `fill_in_contribution_to_jacobian(...)`, rather than `get_residuals()` and `get_jacobian(...)`, respectively. A further advantage of the implementation is that the author need not worry about initialisation of the residuals vector or the Jacobian matrix when using the "`fill_in_`" rather than the "`get_`" functions.

1.5.6.2 Rule 2: Forcing functions, etc. should be implemented as uniquely-named virtual functions

To allow for an interaction between multiple equations, any forcing functions (such as the source function in the Poisson equation) should be implemented as virtual functions. This allows them to be overloaded in derived, multi-physics elements where the forcing function in one equation might depend on the unknowns in another one.
Furthermore, to avoid clashes of function names that may occur when two single-physics elements are combined, member functions that can be expected to have counterparts in the context of other equations should be given suitable modifiers. For instance,
`AdvectionDiffusionEquation::source_fct_adv_diff(...)`
is a better name for a member function that returns the source function in the advection-diffusion equations than
`AdvectionDiffusionEquation::source_fct(...)`
since many other equations are likely to have source functions, too. It is obviously impossible to completely avoid such clashes but this strategy makes them less likely to occur. We are fairly confident that the relevant member functions of all existing elements in `oomph-lib` have been given suitable modifiers to avoid clashes of this type. Therefore, you should be able to combine any element with any other element in the library. If you find a counter-example `let us know`, and we will rectify this in the next release. In the meantime exploit the fact that `oomph-lib` is an open source library; you can change anything you want!

1.5.6.3 Rule 3: Nodal values should only be accessed indirectly via index functions that can be overloaded in derived multi-physics elements

When implementing the single-physics `TwoNodePoissonElement`, discussed above, we provided an access function `u(n)` that returns the (single) nodal value, stored at the element's `n` -th `Node`. The function was implemented as a simple wrapper that followed the pointer to the element's `n` -th `Node` and returned the zero-th nodal value stored at that `Node`:

```
double TwoNodePoissonElement::u(const unsigned &n)
 {
  return node_pt(n)->value(0);
 }
```

In a single-physics context, this implementation is perfectly acceptable since we know *a priori* that in a scalar problem each `Node` only stores a single value.
The same logic suggests that a `TwoNodeAdvectionDiffusionElement` would have a member function

```
double TwoNodeAdvectionDiffusionElement::u(const unsigned &n)
 {
  return node_pt(n)->value(0);
 }
```

However, merging these two elements via multiple inheritance creates two problems. First we have clash of names which could have been avoided by following rule 2 above and calling the two functions `TwoNodePoisson↩Element::u_poisson(...)` and `TwoNodeAdvectionDiffusionElement::u_adv_diff(...)`,

say. More serious is that both elements regard the zero-th nodal value as "their own". This can (and should!) be avoided by the following re-implementation:

```
double TwoNodePoissonElement::u(const unsigned &n)
 {
  return node_pt(n)->value(u_index_poisson());
 }
```

where the virtual function `u_index_poisson()` provides the default index at which the Poisson nodal values are stored in a single-physics context:

```
virtual unsigned TwoNodePoissonElement::u_index_poisson()
 {
  // By default (i.e. in a single-physics context) the Poisson
  // value is stored at the zero-th nodal value:
  return 0;
 }
```

The advection-diffusion element can be modified in the same way:

```
double TwoNodeAdvectionDiffusionElement::u(const unsigned &n)
 {
  return node_pt(n)->value(u_index_adv_diff());
 }
```

with

```
virtual unsigned TwoNodeAdvectionDiffusionElement::u_index_adv_diff()
 {
  // By default (i.e. in a single-physics context) the advection-diffusion
  // value is stored at the zero-th nodal value:
  return 0;
 }
```

In a combined multi-physics problem, we can now merge the two elements by multiple inheritance,

```
class TwoNodeAdvDiffAndPoissonElement :
       public virtual TwoNodePoissonElement,
       public virtual TwoNodeAdvectionDiffusionElement
```

Name clashes are already avoided, so we only have to overwrite the two index functions to indicate which (scalar) value is stored as which nodal value:

```
unsigned TwoNodeAdvDiffAndPoissonElement::u_index_adv_diff()
 {
  // In the combined multi-physics element we continue to store the
  // advection-diffusion value at the zero-th nodal value
  // [as a result it is not actually necessary to re-implement this
  // function!]
  return 0;
 }
```

and

```
unsigned TwoNodeAdvDiffAndPoissonElement::u_index_poisson()
 {
  // In the combined multi-physics element we store the Poisson
  // value at the first nodal value:
  return 1;
 }
```

Specific examples that illustrate the creation of (non-trivial) multi-physics elements by multiple inheritance are provided in the list of example codes.

1.5.6.4 Rule 4: Time derivatives should be implemented in the ALE formulation to allow for moving meshes.

In moving mesh problems, the time-derivative of a nodal value (approximated by the Node's `TimeStepper`) is not the same as the Eulerian time-derivative $\partial/\partial t$ that "usually" occurs in the governing equation. The former represents the rate of change when following the moving node; the latter is the rate of change evaluated at a fixed Eulerian position. The two time-derivatives are related to each other via the ALE relation

$$\left.\frac{\partial(.)}{\partial t}\right|_{Eulerian} = \left.\frac{\partial(.)}{\partial t}\right|_{at\ node} - \sum_{i=1}^{D} v_i^{(Mesh)} \frac{\partial(.)}{\partial x_i}$$

where D is the spatial dimension of the problem and $v_i^{(Mesh)}$ $(i = 1, ..., D)$ is the velocity of the node. Any Eulerian time-derivatives in the governing equation should be implemented in the above form to ensure that the element remains functional in moving mesh problems. See the example describing the solution of the unsteady heat equation in a moving domain for further details.

1.6 PDF file

A pdf version of this document is available. \